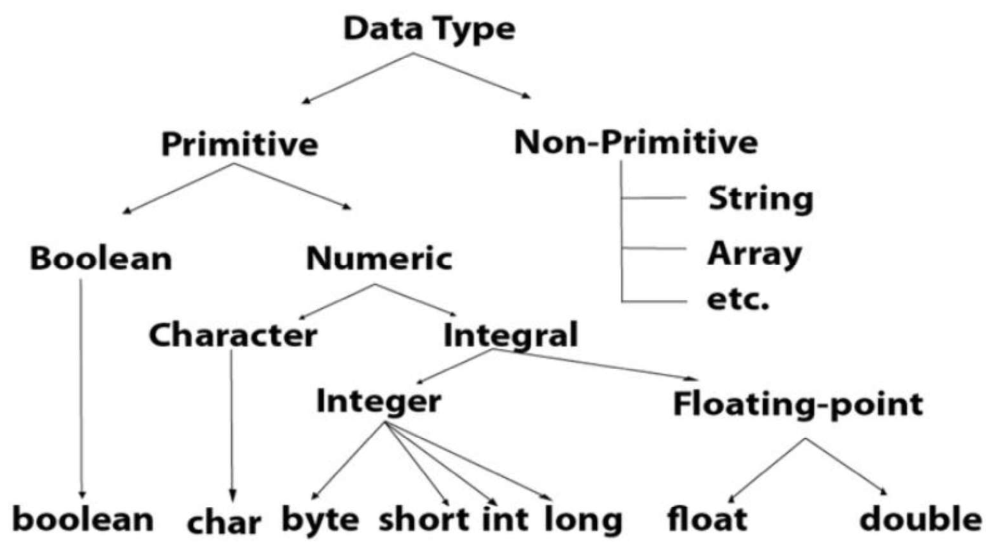
Java Data Types

Java data types define the type of data a variable can store in a program.



Data Type	Default Value	Default size	Example	Range
boolean	false	1 bit	Boolean one = false	true and false
char	'\u0000'	2 byte	char ch = 'A'	'\u0000' (or 0) to '\uffff' (or 65,535) (or) 0 to 65,536
byte	0	1 byte	byte a = 10	-128 to 127
Short	0	2 byte	short s = 10000	–32,768 to 32,767
int	0	4 byte	int a = 100000	–2,147,483,648 to 2,147,483,647
long	0L	8 byte	long a = 100000L	–9,223,372,036,854,775,808 to 9,223,372,036,854,775,807
float	0.0f	4 byte	float f1 = 234.5f	1.4e–045 to 3.4e+038
double	0.0d	8 byte	double d1 = 12.3	4.9e–324 to 1.8e+308

Primitive Data Types

Primitive data types are predefined by Java and are used to store simple values.

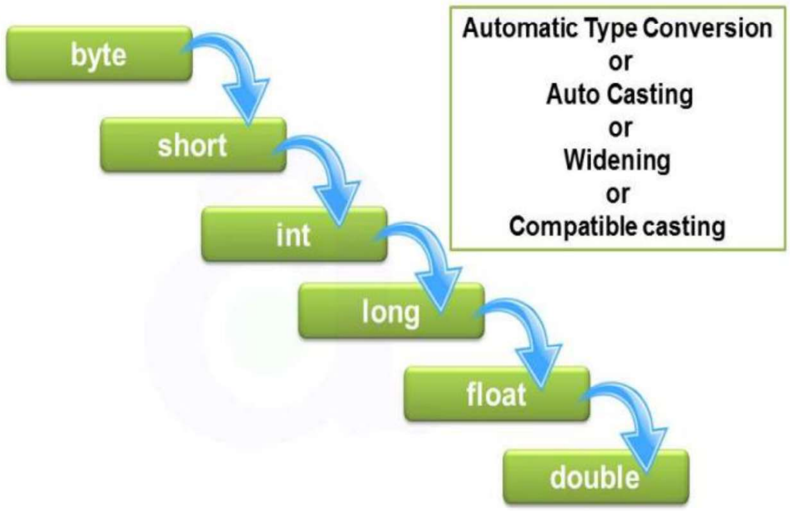
Data Type	Description	Example
byte	Small integer	byte age = 20;
short	Short integer	short year = 2026;
int	Integer	int marks = 95;
long	Large integer	long population = 8000000000L;
float	Single-precision decimal	float price = 99.5**f**;
double	Double-precision decimal	double pi = 3.14159;
char	Single Unicode character	char grade = 'A';
boolean	Logical value	boolean passed = true;

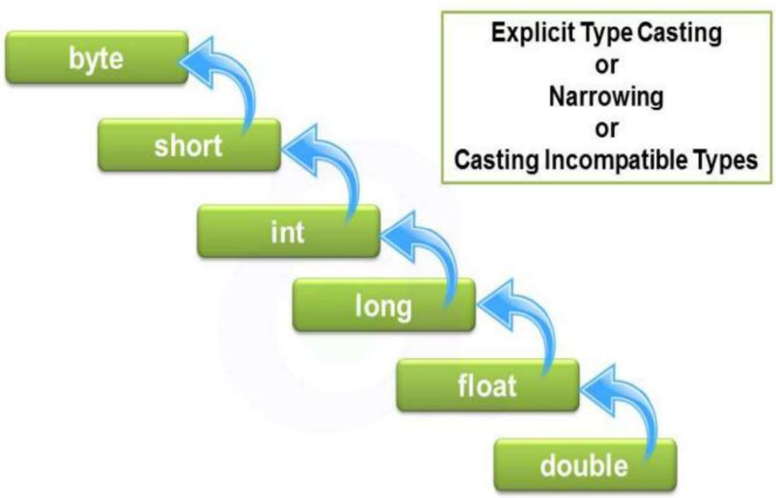
```
public class Main {
    public static void main(String[] args) {
        byte age = 25;
        int salary = 50000;
        System.out.println("Age = " + age);
        System.out.println("Salary = " + salary);
    }
}

Age = 25
Salary = 50000
```

Overflow

```
public class Main {
    public static void main(String[] args) {
        byte b = 127;
        b++;
        System.out.println(b);
    }
}

-128
```

Type Conversion and Casting





Widening (Automatic)

```
public class Main {
    public static void main(String[] args) {
        byte b = 100;
        int i = b;   // Automatic conversion
        System.out.println(i);
    }
}
```

```
100
```

Narrowing (Manual)

```
public class Main {
    public static void main(String[] args) {
        int i = 100;
        byte b = (byte)i;
        System.out.println(b);
    }
}

100
```

Arithmetic with byte: Arithmetic on byte, short, and char always produces an int.

```
public class Main {
    public static void main(String[] args) {

        byte a = 10;
        byte b = 20;

        // byte c = a + b;   // Compilation Error (int + int)

        int c = a + b;

//Type Promotion means Java automatically converts smaller data types to a larger data type before performing an operation.

        System.out.println(c);
    }
}

30
```

Use Explicit Casting

```
public class Main {
    public static void main(String[] args) {
```

```
        byte a = 10;
        byte b = 20;

        byte c = (byte)(a + b);

        System.out.println(c);
    }
}

30

(OR)

public class Main {
    public static void main(String[] args) {

        byte a = 10;
        byte b = 20;

        int c = a + b;

        System.out.println(c);
    }
}

30
```

```
public class Main {
    public static void main(String[] args) {
        char ch1 = 'A';
        char ch2 = 'B';
        //char ch3 = ch1 + ch2;   // Error
        int result = ch1 + ch2;

        System.out.println(result);
    }
}

131
```

Be Careful with Overflow

```
public class Main {
    public static void main(String[] args) {

        byte a = 100;
        byte b = 100;

        byte c = (byte)(a + b);

        System.out.println(c);
    }
}

-56
```

```
// Demonstrate char data type.
class CharDemo {
        public static void main(String args[]) {
                char ch1, ch2;
                ch1 = 88; // code for X
                ch2 = 'Y';
                System.out.print("ch1 and ch2: ");
                System.out.println(ch1 + " " + ch2);
        }
}
output:
ch1 and ch2: X Y
```

```
public class Main {
    public static void main(String[] args) {
        // Constant assignments (Valid)
        byte b = 100;
        short s = 30000;
        char ch = 88;     // Unicode 88 = 'X'
        int i = 50000;
        long l = 100;

        System.out.println("byte  = " + b);
```

```java
        System.out.println("short = " + s);
        System.out.println("char  = " + ch);
        System.out.println("int   = " + i);
        System.out.println("long  = " + l);

        // Widening conversions (Valid)
        byte b1 = 10;

        short s1 = b1;   // byte -> short
        int i1 = s1;     // short -> int
        long l1 = i1;    // int -> long

        System.out.println("\nWidening Conversions:");
        System.out.println("short = " + s1);
        System.out.println("int   = " + i1);
        System.out.println("long  = " + l1);

        // Narrowing conversions
        int x = 100;

        // byte b2 = x;      // ERROR
        // short s2 = x;     // ERROR
        // char ch2 = x;     // ERROR

        // Explicit casting
        byte b2 = (byte) x;
        short s2 = (short) x;
        char ch2 = (char) x;

        System.out.println("\nAfter Explicit Casting:");
        System.out.println("byte  = " + b2);
        System.out.println("short = " + s2);
        System.out.println("char  = " + ch2); // 'd' because Unicode 100 = 'd'
    }
}
```

```java
public class Main {
    public static void main(String[] args) {

        // float initialization
        float f1 = 10;        // int -> float (automatic widening)
        float f2 = 10.5f;     // Correct (float literal)
        float f3 = (float)10.5; // Correct (casting)
```

```
        // double initialization
        double d1 = 10;       // int -> double (automatic widening)
        double d2 = 10.5;     // Correct (double is default)
        double d3 = 10.5f;    // float -> double (automatic widening)

        System.out.println("f1 = " + f1);
        System.out.println("f2 = " + f2);
        System.out.println("f3 = " + f3);

        System.out.println("d1 = " + d1);
        System.out.println("d2 = " + d2);
        System.out.println("d3 = " + d3);

        // Invalid Examples
        // float f4 = 10.5;   // Compilation Error
        // float f5 = d2;     // Compilation Error (double -> float)
    }
}

f1 = 10.0
f2 = 10.5
f3 = 10.5
d1 = 10.0
d2 = 10.5
d3 = 10.5
```

Boolean

```
public class Main {

    // Instance variable (default value)
    boolean defaultValue;

    public static void main(String[] args) {

        // Explicit initialization
        boolean b1 = true;
        boolean b2 = false;

        // Boolean from a comparison
        boolean b3 = (10 > 5);
        boolean b4 = (10 == 20);

        Main obj = new Main();
```

```
        System.out.println("b1 = " + b1);
        System.out.println("b2 = " + b2);
        System.out.println("b3 = " + b3);
        System.out.println("b4 = " + b4);
        System.out.println("Default value = " + obj.defaultValue);
    }
}
```

```
b1 = true
b2 = false
b3 = true
b4 = false
Default value = false
```

```
public class Main {

    boolean defaultValue; //instance variable

    public static void main(String[] args) {

        Main obj = new Main();

        System.out.println(obj.defaultValue);
    }
}
```

```
false
```

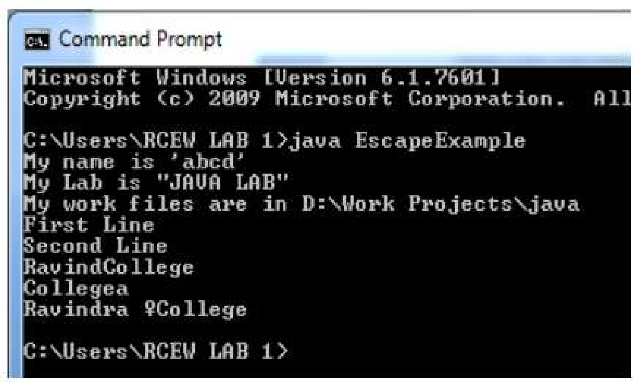
```
public class Main {

    static boolean defaultValue;

    public static void main(String[] args) {

        System.out.println(defaultValue);
    }
}
```

```
False
```

Member	Belongs to	Access from main()
boolean defaultValue;	Object (instance)	☐ Need an object
static boolean defaultValue;	Class	☐ Directly accessible

Character Escape Sequences

```
public class EscapeExample
{
 public static void main(String args[])
 {
        String str = new String ("My name is \'abcd\'");
        System.out.println(str);

        str = "My Lab is \"JAVA LAB\"";
        System.out.println(str);

        str = "My work files are in D:\\Work Projects\\java";
        System.out.println(str);

        str = "First Line \nSecond Line";
        System.out.println(str);

        str = "Ravindra\b\b" + "College";
        System.out.println(str);

        str = "Ravindra \rCollege";
        System.out.println(str);

        str = "Ravindra \fCollege";
        System.out.println(str);
 }
}
```



Variables

The variable is the basic unit of storage in a Java program.

• A variable is defined by the combination of an identifier, a type, and an optional initializer.